

ABDULLAH KHAN

AI/ML Engineer | Agentic AI | RAG Systems | MLOps

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in agentic AI, RAG architectures, and production LLM and voice systems, with 5+ deployed applications handling real user traffic. At CR Equity AI, built and deployed 10 of 25 sub-agents for a CRM AI Assistant platform and am currently building two Twilio/ElevenLabs conversational voice agents. Previously shipped a real-time fraud detection pipeline (Kafka/FastAPI/MLflow) and a multi-cloud security scanner (AWS/Azure/GCP) using LangGraph and the Claude API. Certified by DeepLearning.AI (Stanford) and IBM in Generative AI, RAG, and multi-agent frameworks. BS Computer Science, CGPA 3.59/4.0.

PROFESSIONAL EXPERIENCE

AI Engineer & Platform Agent Development Engineer | July 2026 – Present

CR Equity AI, Inc. | Remote (Tallahassee, FL / Alexandria, VA HQ)

- To surface relevant comps and market context for underwriters, built RAG pipelines and semantic search over real estate and loan data on the proprietary AIVAA™ platform, integrating ChromaDB, Pinecone, and FAISS for valuation comparables and borrower/property profiling.
- Underwriters were manually reading appraisals, rent rolls, T-12 statements, and DSCR packages; automated ingestion and extraction with LLM-powered document workflows, tuning prompt strategies across OpenAI, Claude, and open-source models.
- Built and deployed 10 of 25 sub-agents for the platform's CRM AI Assistant from scratch including Dynamic Pricing, Competitor Monitoring, Compliance & Monitoring, Feedback Loop, Follow-Up Sequences, Performance Analytics, and Task Automation engines as FastAPI microservices on AWS, backed by PostgreSQL (NeonDB), Redis, and Apache Kafka, instrumented with Grafana and OpenTelemetry.
- Architected multi-agent orchestration (LangGraph, LangChain, CrewAI, AutoGen) to automate loan pre-qualification, underwriting routing, and exception flagging.
- The Zoho SalesIQ chatbot wasn't retaining context across turns; rebuilt its conversation flow, knowledge base, and RAG pipeline to fix context retention that same knowledge base now also powers two in-development voice agents. Integrated across Zoho CRM and Zoho Flow to wire AI workflows into the existing CRM stack.
- Currently building two conversational voice agents (Twilio Programmable Voice/Media Streams, ElevenLabs Conversational AI) for the platform: a user-facing technical support agent and a broker-facing sales agent, with FastAPI orchestration and Redis-backed session state.
- Implemented MLOps practices (MLflow, Docker, GitHub Actions CI/CD) for reproducible training and deployment, and built real-time Kafka pipelines for loan event monitoring and fraud detection signals.

iOS Developer Intern | June 2025 – August 2025

Tricode | Faisalabad, Pakistan — On-site

- Assisted in developing and maintaining iOS application features and UI components.
- Worked with the development team to implement, test, debug, and refine application functionality.
- Gained hands-on experience with mobile application development, software development workflows, version control, and debugging.
- Collaborated with team members to understand requirements and translate them into functional application features.

PROJECTS

OneScan: AI-Powered Multi-Cloud Security Scanner

Python · Flask · LangGraph · Claude API · Prowler CLI · Docker · AWS/Azure/GCP

- Raw Prowler CLI output was too technical for non-security stakeholders to act on, so as sole backend developer, built a LangGraph pipeline that turns findings into plain-language risk summaries and remediation guidance via the Claude API.
- Teams needed one scanner instead of three separate cloud tools; unified AWS, Azure, and GCP scanning with platform-specific credential validation (Boto3, Azure Identity SDK, Google Cloud SDK).
- Cloud credentials are a top breach vector, so eliminated plaintext storage entirely, encrypting all credentials with Fernet and decrypting only in-memory during active scans.
- Compliance scans (GDPR, HIPAA, SOC2, CIS) run 2–20 minutes with no visibility into progress; built a Server-Sent Events pipeline streaming live status to the frontend.

Fraud Detection MLOps System

Python · FastAPI · Apache Kafka · MLflow · Docker · Scikit-learn · Gradio

- With fraud at just 0.17% of 284,807 transactions, the default threshold missed too many cases; re-tuned it to 0.265 via precision-recall analysis, cutting false negatives 23%.
- Experiments weren't tracked consistently, making results hard to reproduce; integrated MLflow to log metrics, hyperparameters, and artifacts across every training run.
- Manual setup took hours per deployment; containerized the API, Kafka broker, and Zookeeper with Docker Compose, cutting deployment to minutes.
- Needed the model usable by both engineers and non-technical reviewers; shipped FastAPI endpoints alongside a Gradio UI, live on HuggingFace Spaces with real-time Kafka streaming inference.

UniMind: RAG University Study Assistant

Python · LangChain · ChromaDB · Streamlit · HuggingFace

- Full re-indexing on every new document was slowing retrieval; built an additive ChromaDB pipeline with MMR retrieval that preserves existing chunks, cutting search latency 40%.
- Rate limits on the primary model risked failed sessions; engineered exponential-backoff retries with automatic fallback from GPT-OSS-120B to Qwen2.5-72B-Instruct, reaching 99.9% uptime.
- Students uploaded a mix of scanned and digital PDFs; built an auto-routing OCR pipeline (PyMuPDF + Tesseract) that extracted text from 100% of them.

TECHNICAL SKILLS

Core Stack: Python, LangChain, LangGraph, FastAPI, Docker, MLflow, Apache Kafka, Redis, Git/GitHub, Pydantic

LLM & GenAI: RAG Pipelines, Semantic Search, Vector Databases (ChromaDB, Pinecone, FAISS), Prompt Engineering, AI Agents, LlamaIndex, CrewAI, AutoGen, BeeAI, OpenAI API, Claude API, LLM Fine-tuning (SFT)

Voice & Conversational AI: Twilio (Programmable Voice, Media Streams, SMS), ElevenLabs Conversational AI, Zoho SalesIQ, Zoho Flow, Zoho CRM

ML / Deep Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, Keras, SVM, CNNs, RNNs, Transfer Learning, Hyperparameter Tuning, Anomaly Detection

MLOps & Observability: MLflow, Docker Compose, HuggingFace Spaces, REST APIs, CI/CD (GitHub Actions), Grafana, OpenTelemetry, Prowler CLI

Cloud & Infrastructure: AWS (EC2, S3, IAM), GCP, Azure, PostgreSQL (NeonDB), Fernet Encryption, Cloud Security (CIS, GDPR, HIPAA, SOC2)

Data Engineering: Pandas, NumPy, SQL, Apache Kafka, Web Scraping, EDA, Data Visualization

Collaboration: Jira, Confluence, Slack, Loom, Agile/Scrum

EDUCATION

Bachelor of Science in Computer Science — The University of Faisalabad, Punjab, Pakistan

CGPA: 3.59/4.0 | Completed June 2026, Degree Conferral September 2026

CERTIFICATIONS

- Machine Learning Specialization, DeepLearning.AI & Stanford University
- Deep Learning Specialization, DeepLearning.AI
- IBM RAG and Agentic AI Specialization, IBM
- Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI, IBM
- Generative AI for Data Scientists Specialization, IBM
- Python for Everybody Specialization, University of Michigan